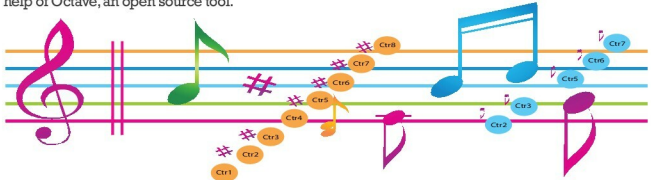


The Imaginary Music of Octave

Here's an introduction to the wonderful world of imaginary numbers, explored with the help of Octave, an open source tool.



Many of us have done powerful matrix math with Octave, without ever thinking about how it works, internally. On that note, let's explore one of the most fascinating, or rather, the more 'imaginary' parts of math.

i Fun

Imaginary numbers are those that don't really exist, but we try to visualise them as points on a 2-D plane with the real part on the x-axis and the imaginary part on the y-axis. So to plot the imaginary i , with a blue star (*), issue the following command at the Octave prompt:

```
$ octave -qf
octave:1> plot(i, "b*")
octave:2>
```

...up pops the display window, as shown in Figure 1.

So what is this i ? It is just the square root of -1. Simple, right? You remember, we got an error trying `sqrt(-1)` in `bc`. Octave is neat. It will answer you politely with an i . If you're still confused, just check out the following:

```
$ octave -qf
octave:1> i
ans = 0 + 1i
octave:2> i * i
ans = -1
octave:3> sqrt(-1)
ans = 0 + 1i
octave:4> i^3
ans = 0 - 1i
octave:5>
```

i Puzzle

And now comes the puzzle part. What is the imaginary part of i^i ? It's zero. What? Is it a real number? Yes it is. It is in fact

equal to $e^{-\pi/2}$. Check it out for yourself:

```
$ octave -qf
octave:1> i^i
ans = 0.20788
octave:2> exp(-pi/2)
ans = 0.20788
octave:3>
```

Yes, you don't really need to bother about how Octave just gets you there so you can have fun with mathematics without getting into it.

i: just a number

As imaginary numbers are just special numbers, they can be used in any operations you may think of with numbers in Octave. Try out `sqrt`, `exp`, `log`, etc—addition, subtraction, multiplication, and division are trivial operations. And you can have even more fun with Octave's power of vectors and matrices. Watch out:

```
$ octave -qf
octave:1> sqrt(i)
ans = 0.70711 + 0.70711i
octave:2> exp(i)
ans = 0.54030 + 0.84147i
octave:3> log(i)
ans = 0.00000 + 1.57080i
octave:4> [i # Press <Enter> here
           i] * [i i]
ans =
   -1  -1
   -1  -1
octave:5>
```

Is $e^{i\theta}$ really $\cos \theta + i \sin \theta$?

Here's a very simple check. Let's plot both the curves. Let θ (theta) be set to values from 0 to π with, say, intervals of